



Advance knowledge.

Drive innovation.

Lead the emerging technology.



MSC IN QUANTUM TECHNOLOGIES BY RESEARCH

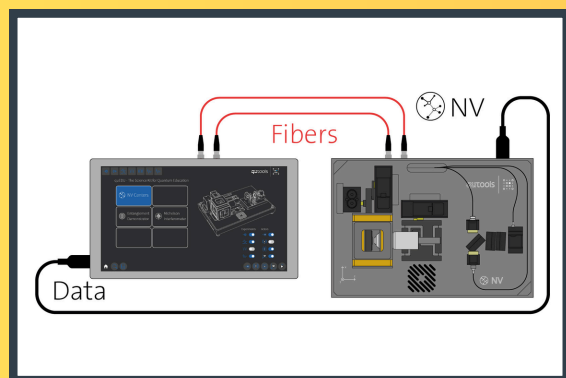
Kengeri Campus, Bengaluru

ABOUT THE PROGRAM

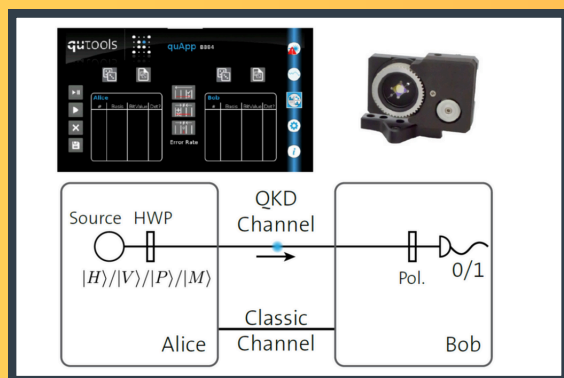
- Quantum technologies are at the forefront of the next technological revolution with transformative applications.
- India's National Quantum Mission (NQM) emphasizes building strong indigenous capabilities in quantum technologies.
- The global quantum technology market is projected to exceed USD 100 billion by 2040, driving demand for quantum-literate engineers, researchers, and developers.
- The MSc in Quantum Technologies (by Research) at CQTCS combines rigorous coursework with intensive, mentored research training.

Shape the Quantum Future?

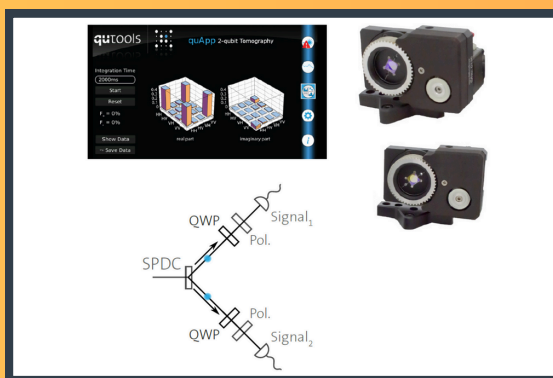
The MSc in Quantum Technologies by Research at CQTCS offers a unique opportunity to be trained at the intersection of fundamental science, cutting-edge technology, and impactful research—preparing you to lead in the next generation of science and technologies in quantum, photonics & semiconductor domains.



Single Photon Experiment Setup



Experiments on Quantum Key Distribution (QKD)



Quantum State Tomography Experiments

Join the Quantum Frontier

The MSc in Quantum Technologies by research at CQTCS offers an intellectually rigorous, research-intensive environment to shape future leaders in cutting-edge science and technology.

Application Portal Link

[https://espro.christuniversity.in/
Application/](https://espro.christuniversity.in/Application/)

Programme Overview

- Programme Name: MSc in Quantum Technologies by Research
- Type: Postgraduate Interdisciplinary Programme
- Department/ Board of Studies (BoS): Centre for Quantum Technologies and Complex Systems (CQTCS)
- School: School of Sciences
- Programme Commencement: June 2026

What Makes This Programme Unique?

- Strong research-driven curriculum from Semester 1
- Advanced training in quantum, photonics, and semiconductor technologies
- Hands-on laboratory experience & mentored research projects
- Seamless pathway to PhD programmes and deep-tech careers





COURSEWORK STRUCTURE

Programme Relevance & Demand

- Quantum technologies represent a critical pillar of future science and engineering. Driven by the National Quantum Mission and global industrial investments, India must cultivate highly trained professionals capable of contributing to quantum, photonics & semiconductor research, product development, and innovation ecosystems.
- By combining advanced coursework, state-of-the-art laboratories, and research-driven training, this programme prepares graduates to seamlessly transition into industry, start-ups, research institutions, and doctoral programmes.

ABOUT THE PROGRAM

Semester 1

- MQT101: Advanced Mathematical Physics
- MQT102: Advanced Classical Mechanics
- MQT103: Quantum Information
- MQT104: Advanced Quantum Mechanics
- MQT105: Advanced Quantum Science Laboratory
- MQT106-RM1: Research Methodology – I
- HED: Holistic Education

Semester 2

- MQT201: Advanced Condensed Matter Physics
- MQT202: Advanced Statistical Mechanics
- MQT203: Photonics Engineering
- MQT204: Quantum Algorithms and Cryptography
- MQT205: Advanced Photonics Laboratory
- MQT206-RM2: Research Methodology – II
- MQT207-RP1: Research Project – I (Topic Selection & Presentation)
- HED: Holistic Education

Semester 3

- MQT301: Quantum Many-body Physics
- MQT302: Semiconductor Chip Fabrication
- MQT303A: Advanced Quantum Information or
- MQT303B: Quantum Field Theory
- MQT304-S: Seminar
- MQT305-RP2: Research Project – II (Progress Presentation & Viva)

Semester 4

- MQT401-RP3: Research Project – III (Thesis, Final Presentation & Viva)

Different bridge courses will be offered to ensure a smooth integration of students from various backgrounds into the programme.

PLACEMENT POTENTIAL

Expected Average Annual Package: ₹7–10 Lakhs

Job Profiles:

- Research Scientist / Research Engineer (in Quantum, Photonics, and Semiconductor industries)
- Quantum Algorithm Developer
- Quantum Application Consultant
- Research Assistant / Research Fellow

Sectors:

- Quantum, Photonics & Semiconductor industries and start-ups
- Research institutions and government laboratories
- IT and advanced technology sectors

Selection Process

Admissions follow the standard process applicable to MSc programmes at CHRIST (Deemed to be University):

- Online Application: Submission of application and academic details through the CHRIST University Portal.
- Personal Interview & Skill Assessment: Shortlisted candidates will be invited to discuss their academic background, research interests, and motivation.
- Final Selection: Based on overall performance in the entrance test and interview, selected candidates will be notified by the Office of Admissions.

Eligibility for Admission

Applicants must possess:

- B.Sc. in Science subjects, or
- M.Sc. in Science subjects, or
- B.Tech / B.E. in Engineering disciplines, or any equivalent qualification.

With a minimum of 60% marks or CGPA 6.5/10.

Candidates must have studied Mathematics as a main or subsidiary subject for at least two years during their qualifying degree.

Higher Education Opportunities

Graduates of this programme will be well-prepared to pursue PhD programmes in Quantum Technologies, Photonics, Semiconductor Fabrication, Complex Systems, and related interdisciplinary fields.

Awards & Recognition

Topper of each batch will be awarded a Gold Medal

